Microsoft Elevate

CAPSTONE PROJECT

GLOBAL SUPERSTORE SALES PERFORMANCE DASHBOARD

PRESENTED BY

STUDENT NAME: SUBHASISH GHOSHAL

COLLEGE NAME: UEM, KOLKATA

DEPARTMENT: CSIT

EMAIL ID: ghoshalsubhasish267@gmail.com

OUTLINE:

- Problem Statement
- Proposed Solution
- System Approach
- Algorithm & Deployment
- Result
- Conclusion
- Future Scope
- References

PROBLEM STATEMENT:

Retail businesses generate a large amount of sales data from different regions, product categories, and customer segments. However, analyzing this raw data manually makes it difficult for organizations to quickly understand their sales performance, profitability trends, and regional distribution of revenue. Decision makers need a clear and interactive way to monitor key business metrics such as total sales, profit, order quantity, and customer segment contribution. Without proper visualization tools, identifying profitable product categories, high-performing regions, and growth trends over time becomes challenging. Therefore, it is important to develop an interactive dashboard that can transform complex sales data into meaningful visual insights. This project focuses on building a Global Superstore Sales Performance Dashboard using Power BI to analyze sales trends, regional performance, customer segmentation, and product profitability, enabling users to explore the data dynamically and support better business decision making.

PROPOSED SOLUTION:

- **Data Collection:**
- Use the Global Superstore dataset containing sales, profit, customer segment, product category, and geographic information.
- **Data Preprocessing:**
- Clean and prepare the dataset by handling missing values, removing unnecessary columns, and correcting data types.
- **Data Visualization:**
  - Create visualizations such as sales by category, sales trend over time, sales by region, profit by sub-category, and sales by customer segment.
- **Interactive Dashboard:**
- Implement filters (Market, Segment, Category, Year) to allow users to interactively explore the data.
- **Deployment:**
- Develop a user-friendly Power BI dashboard that presents business insights in a clear and visually interactive format.

SYSTEM APPROACH:

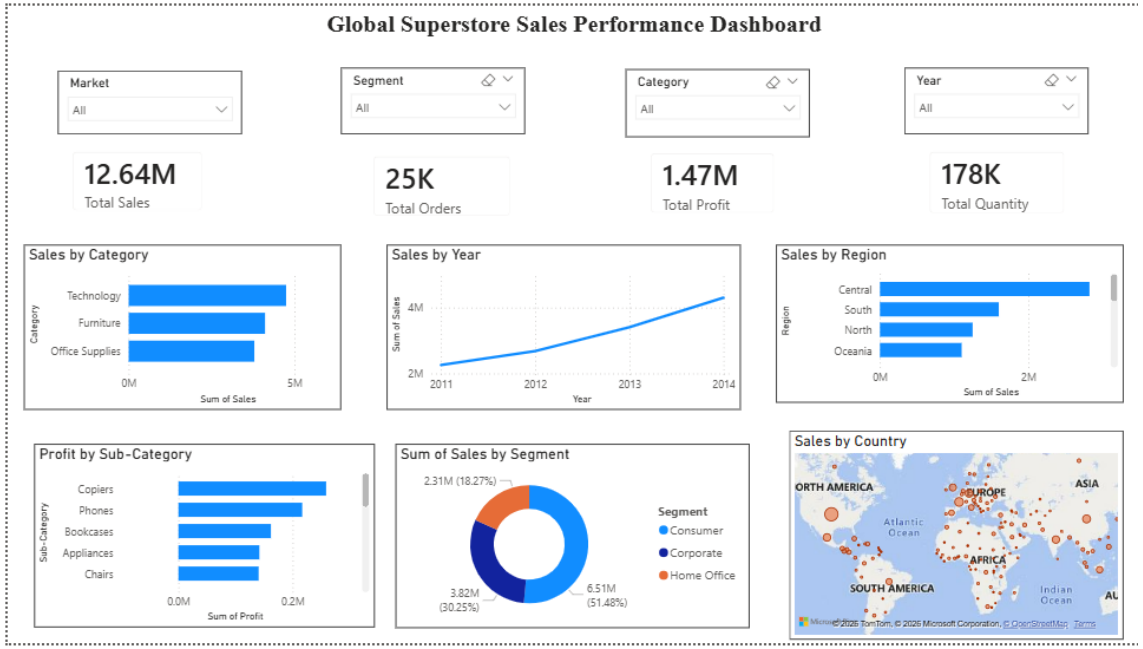
- **System Requirements**
  - Software: Microsoft Power BI Desktop
  - Dataset: Global Superstore Sales Dataset
  - Hardware: Standard computer system capable of running Power BI.
- **Tools and Technologies Used**
  - Power BI for dashboard development and data visualization.
  - Power Query for data cleaning and transformation.
  - DAX (Data Analysis Expressions) for creating calculated measures and KPIs.
- **Visualization Components**
  - Bar Charts – Sales by Category and Sales by Region.
  - Line Chart – Sales Trend over Years.
  - Donut Chart – Sales Distribution by Segment.
  - Map Visualization – Sales by Country.
  - KPI Cards – Total Sales, Total Orders, Total Profit, and Total Quantity.
  - Slicers – Filters for Market, Segment, Category, and Year.

ALGORITHM & DEPLOYMENT:

- **Data Loading:**
  - Import the Global Superstore dataset into Power BI Desktop.
  - Load the dataset into Power Query for further processing.
- **Data Preprocessing:**
  - Clean the dataset by handling missing values and correcting data types.
  - Remove unnecessary fields and prepare the data for analysis.
- **Data Modeling:**
  - Create calculated measures using DAX such as Total Sales, Total Orders, Total Profit, and Total Quantity.
  - Organize the dataset to support efficient analysis and visualization.
- **Dashboard Development:**
  - Create interactive charts and maps to visualize sales performance.
  - Implement slicers for Market, Segment, Category, and Year for dynamic filtering.
- **Deployment:**
  - Combine all visualizations and KPIs into a single dashboard.
  - Publish the dashboard within the Power BI environment for interactive data exploration.

RESULT:

The developed Power BI dashboard successfully analyzes the Global Superstore dataset and presents sales performance through interactive visualizations. Key performance indicators such as total sales, total orders, total profit, and total quantity provide a quick overview of business performance. The visualizations highlight sales trends over time, regional sales distribution, category-wise sales, and profit by sub-category. Interactive slicers allow users to filter the data by market, segment, category, and year, enabling dynamic exploration of the dataset and better understanding of sales patterns.



CONCLUSION:

The project successfully developed an interactive Power BI dashboard to analyze the Global Superstore dataset and visualize sales performance. The dashboard provides clear insights into key business metrics such as total sales, profit, orders, and quantity through various visualizations. Interactive filters and charts allow users to explore sales trends, regional performance, and product category distribution effectively. Overall, the dashboard helps transform raw data into meaningful insights that support better business analysis and decision-making.

FUTURE SCOPE:

In the future, the dashboard can be enhanced by integrating additional datasets to provide deeper business insights. Advanced analytics and predictive models can be incorporated to forecast future sales trends and demand patterns. The dashboard can also be connected to real-time data sources to enable live monitoring of sales performance. Further improvements may include adding more interactive visualizations and deploying the dashboard on Power BI Service for broader accessibility and collaboration.

REFERENCES:

- Global Superstore Dataset – Kaggle
Global Super Store Dataset (Kaggle)
- Microsoft Power BI Documentation
https://learn.microsoft.com/power-bi/
- GitHub(Link)

Thank You